Richard Lin

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EDUCATION

Temple University *Philadelphia, PA*
Bachelor of Science in Electrical and Computer Engineering *Expected Graduation: May 2028*
Minor: Computer Science | Fundamentals of Physics Certificate | GPA: 3.88/4.00 | Dean's List: F24, S25, F25
Courses: Classical Control Systems, Signals & Systems, Processor Systems, Engineering Computation IV (GPU Programming)

PROFESSIONAL EXPERIENCE

Undergraduate Research Assistant *Oct 2025 – Present*
Temple University – Computer Fusion Lab (CFL) *Philadelphia, PA*
Advisor: Dr. Li Bai

- Administered lab systems and website, handling upgrades, troubleshooting, security, and network reliability
- Built a Python multi-agent LLM system with Temple's TechOWL to match users' natural-language needs to the top 3 assistive devices for people with disabilities, using RAG + LLM reranking on a daily-synced 400+ item catalog
- Trained an ACT model with LeRobot for assistive robotic-arm pickup and handover; developed ROS-based C++ motion control and integrated STT/TTS for Mini Pupper and the ROBOTIS OP3 humanoid

Undergraduate Teaching Assistant *Aug 2025 – Present*
Temple University *Philadelphia, PA*

- Assisted faculty with instruction, hands-on activities, and assignment evaluation; held office hours for student support
- **Intro to Engineering:** Guided students building embedded Micro:bit incubator, testing components (LEDs, fans, sensors)
- **Engineering Problem Solving:** Supported wind turbine calculation/data analysis using Excel, Python (NumPy, Matplotlib)
- **Engineering Computation I:** Taught C/C++/Python/Linux/Bash fundamentals, plus basic algorithms and time complexity

STEPS Ambassador *Sep 2024 – Present*
Temple University – Sustainable Temple Energy and Power Scholars (STEPS) *Philadelphia, PA*

- Mentored STEPS underclassmen through the high-school-to-college transition, providing academic guidance
- Designed and led renewable-energy STEM demos for high schoolers, emphasizing sustainability and real-world engineering

PROJECTS

Musical Light Show – ECE 2352: Circuits and Electronics II *Apr 2026 – May 2026*

- Designed and built an analog audio-processing system that combines left/right audio channels, and separates high and low frequency signals to drive an audio-reactive LED light show
- Implemented dual op-amp stages for summing and volume/gain control using a feedback potentiometer, high/low-pass filters, peak detectors, BJT LED driver circuits
- Calculated values from gains and cutoff frequency using Ohm's Law and KVL; verified in Multisim and hardware within 1%

Detective Conan AI Glasses (Childhood Dream Project) *Aug 2026 – Present*

- Architecting smart glasses on Pi Zero 2 W with camera, mic, solar charging, 3D-printed enclosure; Google Gemini API
- Developing software architecture with intent, LLM routing, object detection, OCR, face recognition, and cybersecurity

CLUBS & LEADERSHIP

President *Sep 2024 – Present*
Institute of Electrical and Electronics Engineers (IEEE) – Temple University *Philadelphia, PA*

- Led 14-member IEEE executive board in planning outreach programs, technical workshops, and industry networking events, while coordinating with four affiliated chapters and societies (WIE, PES, EMBS, HKN)
- Grew IEEE SAC participation by 50% (14 → 21) through targeted event promotion and outreach during 2025–2026
- Founded the first annual multi-university hardware hackathon with Drexel and UPenn IEEE branches

Vice President *Jun 2026 – Present*
Eta Kappa Nu (IEEE-HKN), Iota Sigma Chapter – Temple University *Philadelphia, PA*

SKILLS

Programming: Python, C, C++, Java, MATLAB, Bash, SystemVerilog, JavaScript, HTML/CSS, CUDA, AVR Assembly
AI & Robotics: Machine Learning, PyTorch, LLM Integration, RAG, VLA, Edge AI, Computer Vision (OpenCV, YOLO), ROS
Hardware & Tools: Git/GitHub, Linux, Raspberry Pi, Arduino, FPGA, Circuit Design & Analysis, Soldering, n8n, PCB Design
Languages: English, Mandarin Chinese, Fuzhounese, Spanish